

```

28  const payload = {
29      ...data,
30      createdAt: serverTimestamp(),
31      status: 'RECYCLED',
32  };
33  const ref = await addDoc(
34      collection(db, 'wasteProcessing'), payload
35  );
36
37  if (data.intakeId) {
38      const intakeRef = doc(db, 'wasteIntakes', data.intakeId);
39      await updateDoc(intakeRef, {
40          status: 'PROCESSED',
41          lastUpdate: serverTimestamp(),
42      });
43  }
44  return ref.id;
45  }

```

Listing 8.14: Waste Delivery Functions (firebase-service.ts)

These functions implement a cascading update pattern where recording a delivery or processing event automatically updates the parent document's status. This ensures data consistency across the waste management pipeline without requiring client-side orchestration.

8.14 Listing Status Workflow Functions

The following functions implement the complete listing status transition workflow:

```

1  export async function startPickup(
2      listingId: string
3  ): Promise<void> {
4      const ref = doc(db, 'listings', listingId);
5      await updateDoc(ref, {
6          status: 'IN_TRANSIT' as ListingStatus,
7          updatedAt: serverTimestamp(),
8      });
9  }
10
11 export async function markDelivered(
12     listingId: string
13 ): Promise<void> {
14     const ref = doc(db, 'listings', listingId);
15     await updateDoc(ref, {

```